

BannerFlow Manager API

1.0

Generated by Doxygen 1.9.8

Chapter 1

README

BannerFlow Manager

Veb-prilozhenie dlya upravleniya reklamnymi bannerami v igrah s generatsiey koda dlya Godot Engine.

Bystryy start

Aktivatsiya virtualnogo okruzheniya

source myenv/bin/activate

Ustanovka zavisimostey

pip install -r requirements.txt

Nastroyka bazy dannyh

python [manage.py](#) migrate

Zapusk servera

python [manage.py](#) runserver

Otkroyte v brauzere: <http://127.0.0.1:8000/>

Struktura proekta

bannerflow_app/ - Osnovnoe prilozhenie Django

├─ migrations/ - Migratsii BD

├─ [models.py](#) - Modeli dannyh

├─ [views.py](#) - Kontrollery

├─ [services.py](#) - Biznes-logika

├─ [banner_generator.py](#) - Generator bannerov

└─ data_parser.py - Parser CSV/XLSX

bannerflow_project/ - Nastroyki proekta

├─ [settings.py](#) - Konfiguratsiya

└─ urls.py - URL marshruty

docs/ - Dokumentatsiya

├─ html/ - Doxygen HTML

└─ pdf_output/ - PDF dokumentatsiya

media/ - Mediafayly
|— banners/ - Sgenerirovannye bannery
|— exports/godot/ - Skripty dlya Godot

.gitignore - Ignoriruemye fayly
requirements.txt - Zavisimosti Python
[manage.py](#) - Django skript
pytest.ini - Konfiguratsiya testov
doxygen.config - Konfiguratsiya Doxygen

Osnovnye vozmozhnosti
Parsing igrovyyh dannyh - zagruzka CSV/XLSX faylov
Generatsiya bannerov - sozдание reklamnyh bannerov
Eksport v Godot - avtomaticheskaya generatsiya .gd skriptov
A/B testirovanie - sravnenie effektivnosti bannerov
Polnaya dokumentatsiya - Doxygen + LaTeX

Modeli dannyh
Campaign - Reklamnye kampanii
Banner - Bannery
BannerTemplate - Shablony bannerov
GameData - Igrovye dannye
ABTest - A/B testy
GeneratedBanner - Sgenerirovannye bannery

Testirovanie
Zapusk vseh testov
./run_tests.sh

Zapusk s pokrytiem
pytest --cov=[bannerflow_app](#) tests/

Dokumentatsiya
Generatsiya HTML dokumentatsii
doxygen doxygen.config
Fayly dostupny v docs/html/

Tehnologii
Python 3.13 + Django 5.2.8
SQLite - baza dannyh
Pillow - obrabotka izobrazheniy
pandas - analiz dannyh
Doxygen - dokumentatsiya
pytest - testirovanie

UML diagrammy
Use Case diagramma

Deployment diagramma

Chapter 2

Namespace Index

2.1 Namespace List

Here is a list of all namespaces with brief descriptions:

bannerflow_app	??
bannerflow_app.admin	??
bannerflow_app.banner_generator	??
bannerflow_app.models	??
bannerflow_app.services	??
bannerflow_app.tests	??
bannerflow_app.urls	??
bannerflow_app.views	??
bannerflow_project	??
bannerflow_project.settings	??
bannerflow_project.urls	??
bannerflow_project.wsgi	??
manage	??

Chapter 3

Hierarchical Index

3.1 Class Hierarchy

This inheritance list is sorted roughly, but not completely, alphabetically:

bannerflow_app.banner_generator.BannerGenerator	??
bannerflow_app.services.BannerService	??
bannerflow_app.models.BannerTemplate.Meta	??
bannerflow_app.models.CampaignStatistics.Meta	??
bannerflow_app.models.GameData.Meta	??
bannerflow_app.models.GeneratedBanner.Meta	??
models.Model	
bannerflow_app.models.ABTest	??
bannerflow_app.models.Banner	??
bannerflow_app.models.BannerTemplate	??
bannerflow_app.models.Campaign	??
bannerflow_app.models.CampaignStatistics	??
bannerflow_app.models.GameData	??
bannerflow_app.models.GeneratedBanner	??
bannerflow_app.models.UserProfile	??
admin.ModelAdmin	
bannerflow_app.admin.ABTestAdmin	??
bannerflow_app.admin.BannerAdmin	??
bannerflow_app.admin.BannerTemplateAdmin	??
bannerflow_app.admin.CampaignAdmin	??
bannerflow_app.admin.CampaignStatisticsAdmin	??
bannerflow_app.admin.GameDataAdmin	??
bannerflow_app.admin.GeneratedBannerAdmin	??
bannerflow_app.admin.UserProfileAdmin	??
bannerflow_app.services.NotificationService	??
admin.StackedInline	
bannerflow_app.admin.UserProfileInline	??
bannerflow_app.services.StatisticsService	??
TestCase	
bannerflow_app.tests.TestAPIAndServices	??
bannerflow_app.tests.TestBannerGenerator	??
bannerflow_app.tests.TestBannerModel	??
bannerflow_app.tests.TestBannerTemplateModel	??
bannerflow_app.tests.TestCampaignModel	??
bannerflow_app.tests.TestGameDataModel	??
bannerflow_app.tests.TestNegativeScenarios	??
UserAdmin	
bannerflow_app.admin.CustomUserAdmin	??

Chapter 4

Class Index

4.1 Class List

Here are the classes, structs, unions and interfaces with brief descriptions:

bannerflow_app.models.ABTest	??
bannerflow_app.admin.ABTestAdmin	??
bannerflow_app.models.Banner	??
bannerflow_app.admin.BannerAdmin	??
bannerflow_app.banner_generator.BannerGenerator	??
bannerflow_app.services.BannerService	??
bannerflow_app.models.BannerTemplate	??
bannerflow_app.admin.BannerTemplateAdmin	??
bannerflow_app.models.Campaign	??
bannerflow_app.admin.CampaignAdmin	??
bannerflow_app.models.CampaignStatistics	??
bannerflow_app.admin.CampaignStatisticsAdmin	??
bannerflow_app.admin.CustomUserAdmin	??
bannerflow_app.models.GameData	??
bannerflow_app.admin.GameDataAdmin	??
bannerflow_app.models.GeneratedBanner	??
bannerflow_app.admin.GeneratedBannerAdmin	??
bannerflow_app.models.BannerTemplate.Meta	??
bannerflow_app.models.CampaignStatistics.Meta	??
bannerflow_app.models.GameData.Meta	??
bannerflow_app.models.GeneratedBanner.Meta	??
bannerflow_app.services.NotificationService	??
bannerflow_app.services.StatisticsService	??
bannerflow_app.tests.TestAPIAndServices	??
bannerflow_app.tests.TestBannerGenerator	??
bannerflow_app.tests.TestBannerModel	??
bannerflow_app.tests.TestBannerTemplateModel	??
bannerflow_app.tests.TestCampaignModel	??
bannerflow_app.tests.TestGameDataModel	??
bannerflow_app.tests.TestNegativeScenarios	??
bannerflow_app.models.UserProfile	??
bannerflow_app.admin.UserProfileAdmin	??
bannerflow_app.admin.UserProfileInline	??

Chapter 5

File Index

5.1 File List

Here is a list of all files with brief descriptions:

manage.py	??
bannerflow_app/ __init__.py	??
bannerflow_app/ admin.py	??
bannerflow_app/ banner_generator.py	??
bannerflow_app/ models.py	??
bannerflow_app/ services.py	??
bannerflow_app/ tests.py	??
bannerflow_app/ urls.py	??
bannerflow_app/ views.py	??
bannerflow_project/ __init__.py	??
bannerflow_project/ settings.py	??
bannerflow_project/ urls.py	??
bannerflow_project/ wsgi.py	??

Chapter 6

Namespace Documentation

6.1 bannerflow_app Namespace Reference

Namespaces

- namespace [admin](#)
- namespace [banner_generator](#)
- namespace [models](#)
- namespace [services](#)
- namespace [tests](#)
- namespace [urls](#)
- namespace [views](#)

6.1.1 Detailed Description

Created on 9 noyab. 2025 g.

@author: dorokhova

6.2 bannerflow_app.admin Namespace Reference

Classes

- class [ABTestAdmin](#)
- class [BannerAdmin](#)
- class [BannerTemplateAdmin](#)
- class [CampaignAdmin](#)
- class [CampaignStatisticsAdmin](#)
- class [CustomUserAdmin](#)
- class [GameDataAdmin](#)
- class [GeneratedBannerAdmin](#)
- class [UserProfileAdmin](#)
- class [UserProfileInline](#)

6.3 bannerflow_app.banner_generator Namespace Reference

Classes

- class [BannerGenerator](#)

6.4 bannerflow_app.models Namespace Reference

Classes

- class [ABTest](#)
- class [Banner](#)
- class [BannerTemplate](#)
- class [Campaign](#)
- class [CampaignStatistics](#)
- class [GameData](#)
- class [GeneratedBanner](#)
- class [UserProfile](#)

Functions

- [validate_media_file](#) (value)

6.4.1 Function Documentation

6.4.1.1 validate_media_file()

```
bannerflow_app.models.validate_media_file (  
    value )
```

Validator dlya mediafaylov - tolko PNG, JPG, MP4

6.5 bannerflow_app.services Namespace Reference

Classes

- class [BannerService](#)
- class [NotificationService](#)
- class [StatisticsService](#)

6.6 bannerflow_app.tests Namespace Reference

Classes

- class [TestAPIAndServices](#)
- class [TestBannerGenerator](#)
- class [TestBannerModel](#)
- class [TestBannerTemplateModel](#)
- class [TestCampaignModel](#)
- class [TestGameDataModel](#)
- class [TestNegativeScenarios](#)

6.6.1 Detailed Description

Modul testirovaniya dlya BannerFlow Manager

6.7 bannerflow_app.urls Namespace Reference

Variables

- list [urlpatterns](#)

6.7.1 Variable Documentation

6.7.1.1 urlpatterns

list bannerflow_app.urls.urlpatterns

Initial value:

```
00001 = [  
00002     # Osnovnye stranitsy  
00003     path("", views.home, name='home'),  
00004     path('dashboard/', views.dashboard, name='dashboard'),  
00005  
00006     # Generator bannerov (osnovnaya funktsionalnost proekta)  
00007     path('generator/', views.banner_generator, name='banner_generator'),  
00008     path('generate/', views.generate_banner, name='generate_banner'),  
00009     path('banner/preview/<int:banner_id>', views.banner_preview, name='banner_preview'),  
00010     path('banner/export/<int:banner_id>', views.export_banner, name='export_banner'),  
00011     path('banner/list/', views.banner_list, name='banner_list'),  
00012  
00013     # API  
00014     path('api/public/banners/', views.api_banners, name='public_banners'),  
00015  
00016     path('upload-data/', views.upload_game_data, name='upload_game_data'),  
00017 ]
```

6.8 bannerflow_app.views Namespace Reference

Functions

- [home](#) (request)
- [dashboard](#) (request)
- [banner_generator](#) (request)
- [generate_banner](#) (request)
- [banner_preview](#) (request, banner_id)
- [api_banners](#) (request)
- [upload_game_data](#) (request)
- [export_banner](#) (request, banner_id)
- [banner_list](#) (request)

6.8.1 Function Documentation

6.8.1.1 `api_banners()`

```
bannerflow_app.views.api_banners (
    request )
```

API dlya polucheniya bannerov

6.8.1.2 `banner_generator()`

```
bannerflow_app.views.banner_generator (
    request )
```

Glavnaya stranitsa generatora bannerov

6.8.1.3 `banner_list()`

```
bannerflow_app.views.banner_list (
    request )
```

Spisok bannerov dlya eksporta

6.8.1.4 `banner_preview()`

```
bannerflow_app.views.banner_preview (
    request,
    banner_id )
```

Predprosmotr sgenerirovannogo bannera

6.8.1.5 `dashboard()`

```
bannerflow_app.views.dashboard (
    request )
```

Panel upravleniya

6.8.1.6 export_banner()

```
bannerflow_app.views.export_banner (
    request,
    banner_id )
```

Eksport bannera dlya Godot - RABOChAYa VERSIYa

6.8.1.7 generate_banner()

```
bannerflow_app.views.generate_banner (
    request )
```

Generatsiya bannera

6.8.1.8 home()

```
bannerflow_app.views.home (
    request )
```

Domashnyaya stranitsa

6.8.1.9 upload_game_data()

```
bannerflow_app.views.upload_game_data (
    request )
```

Zagruzka dannyh igry

6.9 bannerflow_project Namespace Reference

Namespaces

- namespace [settings](#)
- namespace [urls](#)
- namespace [wsgi](#)

6.9.1 Detailed Description

Created on 9 noyab. 2025 g.

@author: dorokhova

6.10 bannerflow_project.settings Namespace Reference

Variables

- `BASE_DIR` = `Path(__file__).resolve().parent.parent`
- `str SECRET_KEY` = `'django-insecure-your-secret-key-here'`
- `bool DEBUG` = `True`
- `list ALLOWED_HOSTS` = `[]`
- `list INSTALLED_APPS`
- `list MIDDLEWARE`
- `str ROOT_URLCONF` = `'bannerflow_project.urls'`
- `list TEMPLATES`
- `str WSGI_APPLICATION` = `'bannerflow_project.wsgi.application'`
- `dict DATABASES`
- `str LANGUAGE_CODE` = `'en-us'`
- `str TIME_ZONE` = `'UTC'`
- `bool USE_I18N` = `True`
- `bool USE_TZ` = `True`
- `str STATIC_URL` = `'static/'`
- `str DEFAULT_AUTO_FIELD` = `'django.db.models.BigAutoField'`
- `dict REST_FRAMEWORK`
- `str MEDIA_URL` = `'/media/'`
- `MEDIA_ROOT` = `os.path.join(BASE_DIR, 'media')`
- `str EMAIL_BACKEND` = `'django.core.mail.backends.console.EmailBackend'`
- `str DEFAULT_FROM_EMAIL` = `'noreply@bannerflow.com'`
- `str EMAIL_HOST` = `'smtp.mail.ru'`
- `int EMAIL_PORT` = `465`
- `bool EMAIL_USE_SSL` = `True`
- `str EMAIL_HOST_USER` = `'bannerflow_manager@mail.ru'`
- `str EMAIL_HOST_PASSWORD` = `'DyAQzqEBGdfyyBBI3iym'`
- `str SERVER_EMAIL` = `'bannerflow_manager@mail.ru'`

6.10.1 Detailed Description

Django settings for bannerflow_project.

6.10.2 Variable Documentation

6.10.2.1 ALLOWED_HOSTS

```
list bannerflow_project.settings.ALLOWED_HOSTS = []
```

6.10.2.2 BASE_DIR

```
bannerflow_project.settings.BASE_DIR = Path(__file__).resolve().parent.parent
```

6.10.2.3 DATABASES

```
dict bannerflow_project.settings.DATABASES
```

Initial value:

```
00001 = {
00002     'default': {
00003         'ENGINE': 'django.db.backends.sqlite3',
00004         'NAME': BASE_DIR / 'db.sqlite3',
00005     }
00006 }
```

6.10.2.4 DEBUG

```
bool bannerflow_project.settings.DEBUG = True
```

6.10.2.5 DEFAULT_AUTO_FIELD

```
str bannerflow_project.settings.DEFAULT_AUTO_FIELD = 'django.db.models.BigAutoField'
```

6.10.2.6 DEFAULT_FROM_EMAIL

```
str bannerflow_project.settings.DEFAULT_FROM_EMAIL = 'noreply@bannerflow.com'
```

6.10.2.7 EMAIL_BACKEND

```
str bannerflow_project.settings.EMAIL_BACKEND = 'django.core.mail.backends.console.Email↵
Backend'
```

6.10.2.8 EMAIL_HOST

```
str bannerflow_project.settings.EMAIL_HOST = 'smtp.mail.ru'
```

6.10.2.9 EMAIL_HOST_PASSWORD

```
str bannerflow_project.settings.EMAIL_HOST_PASSWORD = 'DyAQzqEBGdfyyBB13iyM'
```

6.10.2.10 EMAIL_HOST_USER

```
str bannerflow_project.settings.EMAIL_HOST_USER = 'bannerflow_manager@mail.ru'
```

6.10.2.11 EMAIL_PORT

```
int bannerflow_project.settings.EMAIL_PORT = 465
```

6.10.2.12 EMAIL_USE_SSL

```
bool bannerflow_project.settings.EMAIL_USE_SSL = True
```

6.10.2.13 INSTALLED_APPS

```
list bannerflow_project.settings.INSTALLED_APPS
```

Initial value:

```
00001 = [  
00002     'django.contrib.admin',  
00003     'django.contrib.auth',  
00004     'django.contrib.contenttypes',  
00005     'django.contrib.sessions',  
00006     'django.contrib.messages',  
00007     'django.contrib.staticfiles',  
00008     'rest_framework',  
00009     'bannerflow_app',  
00010 ]
```

6.10.2.14 LANGUAGE_CODE

```
str bannerflow_project.settings.LANGUAGE_CODE = 'en-us'
```

6.10.2.15 MEDIA_ROOT

```
bannerflow_project.settings.MEDIA_ROOT = os.path.join(BASE_DIR, 'media')
```

6.10.2.16 MEDIA_URL

```
str bannerflow_project.settings.MEDIA_URL = '/media/'
```

6.10.2.17 MIDDLEWARE

```
list bannerflow_project.settings.MIDDLEWARE
```

Initial value:

```
00001 = [  
00002     'django.middleware.security.SecurityMiddleware',  
00003     'django.contrib.sessions.middleware.SessionMiddleware',  
00004     'django.middleware.common.CommonMiddleware',  
00005     'django.middleware.csrf.CsrfViewMiddleware',  
00006     'django.contrib.auth.middleware.AuthenticationMiddleware',  
00007     'django.contrib.messages.middleware.MessageMiddleware',  
00008     'django.middleware.clickjacking.XFrameOptionsMiddleware',  
00009 ]
```

6.10.2.18 REST_FRAMEWORK

```
dict bannerflow_project.settings.REST_FRAMEWORK
```

Initial value:

```
00001 = {  
00002     'DEFAULT_RENDERER_CLASSES': [  
00003         'rest_framework.renderers.JSONRenderer', # Tolko JSON, bez HTML  
00004     ],  
00005     'DEFAULT_PERMISSION_CLASSES': [  
00006         'rest_framework.permissions.IsAuthenticated',  
00007     ]  
00008 }
```

6.10.2.19 ROOT_URLCONF

```
str bannerflow_project.settings.ROOT_URLCONF = 'bannerflow_project.urls'
```

6.10.2.20 SECRET_KEY

```
str bannerflow_project.settings.SECRET_KEY = 'django-insecure-your-secret-key-here'
```

6.10.2.21 SERVER_EMAIL

```
str bannerflow_project.settings.SERVER_EMAIL = 'bannerflow_manager@mail.ru'
```

6.10.2.22 STATIC_URL

```
str bannerflow_project.settings.STATIC_URL = 'static/'
```

6.10.2.23 TEMPLATES

```
list bannerflow_project.settings.TEMPLATES
```

Initial value:

```
00001 = [  
00002     {  
00003         'BACKEND': 'django.template.backends.django.DjangoTemplates',  
00004         'DIRS': [],  
00005         'APP_DIRS': True,  
00006         'OPTIONS': {  
00007             'context_processors': [  
00008                 'django.template.context_processors.debug',  
00009                 'django.template.context_processors.request',  
00010                 'django.contrib.auth.context_processors.auth',  
00011                 'django.contrib.messages.context_processors.messages',  
00012             ],  
00013         },  
00014     ],  
00015 ]
```

6.10.2.24 TIME_ZONE

```
str bannerflow_project.settings.TIME_ZONE = 'UTC'
```

6.10.2.25 USE_I18N

```
bool bannerflow_project.settings.USE_I18N = True
```

6.10.2.26 USE_TZ

```
bool bannerflow_project.settings.USE_TZ = True
```

6.10.2.27 WSGI_APPLICATION

```
str bannerflow_project.settings.WSGI_APPLICATION = 'bannerflow_project.wsgi.application'
```

6.11 bannerflow_project.urls Namespace Reference

Variables

- list [urlpatterns](#)
- [MEDIA_URL](#)
- [document_root](#)

6.11.1 Variable Documentation

6.11.1.1 document_root

```
bannerflow_project.urls.document_root
```

6.11.1.2 MEDIA_URL

```
bannerflow_project.urls.MEDIA_URL
```

6.11.1.3 urlpatterns

```
list bannerflow_project.urls.urlpatterns
```

Initial value:

```
00001 = [  
00002     path('admin/', admin.site.urls),  
00003     path("", include('bannerflow_app.urls')),  
00004 ]
```

6.12 bannerflow_project.wsgi Namespace Reference

Variables

- [application](#) = `get_wsgi_application()`

6.12.1 Detailed Description

```
WSGI config for bannerflow_project project.
```

6.12.2 Variable Documentation

6.12.2.1 application

```
bannerflow_project.wsgi.application = get_wsgi_application()
```

6.13 manage Namespace Reference

Functions

- [main\(\)](#)

6.13.1 Detailed Description

Django's command-line utility for administrative tasks.

6.13.2 Function Documentation

6.13.2.1 main()

```
manage.main ( )
```

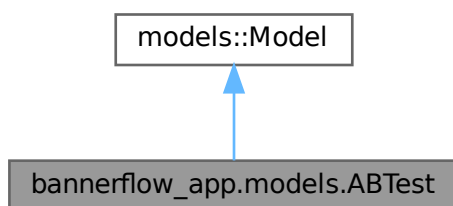
Run administrative tasks.

Chapter 7

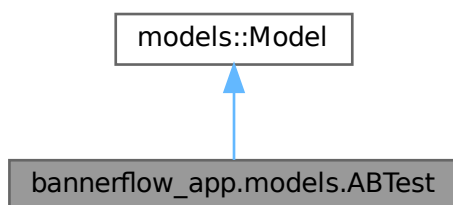
Class Documentation

7.1 bannerflow_app.models.ABTest Class Reference

Inheritance diagram for bannerflow_app.models.ABTest:



Collaboration diagram for bannerflow_app.models.ABTest:



Public Member Functions

- [__str__](#) (self)

Static Public Attributes

- `name` = `models.CharField(max_length=255, verbose_name="Nazvanie testa")`
- `campaign` = `models.ForeignKey(Campaign, on_delete=models.CASCADE, verbose_name="Kampaniya")`
- `banners` = `models.ManyToManyField(Banner, verbose_name="Bannery dlya testa")`
- `audience_segment` = `models.CharField(max_length=100, verbose_name="Segment auditorii")`
- `start_date` = `models.DateTimeField(default=timezone.now, verbose_name="Data nachala")`
- `end_date` = `models.DateTimeField(default=timezone.now, verbose_name="Data okonchaniya")`
- `is_active` = `models.BooleanField(default=True, verbose_name="Aktiven")`
- `created_at` = `models.DateTimeField(auto_now_add=True)`

7.1.1 Member Function Documentation

7.1.1.1 `__str__()`

```
bannerflow_app.models.ABTest.__str__ (
    self )
```

7.1.2 Member Data Documentation

7.1.2.1 `audience_segment`

```
bannerflow_app.models.ABTest.audience_segment = models.CharField(max_length=100, verbose_name="Segment auditorii") [static]
```

7.1.2.2 `banners`

```
bannerflow_app.models.ABTest.banners = models.ManyToManyField(Banner, verbose_name="Bannery dlya testa") [static]
```

7.1.2.3 `campaign`

```
bannerflow_app.models.ABTest.campaign = models.ForeignKey(Campaign, on_delete=models.CASCADE, verbose_name="Kampaniya") [static]
```

7.1.2.4 `created_at`

```
bannerflow_app.models.ABTest.created_at = models.DateTimeField(auto_now_add=True) [static]
```

7.1.2.5 `end_date`

```
bannerflow_app.models.ABTest.end_date = models.DateTimeField(default=timezone.now, verbose_name="Data okonchaniya") [static]
```

7.1.2.6 is_active

```
bannerflow_app.models.ABTest.is_active = models.BooleanField(default=True, verbose_name="Aktiven")  
[static]
```

7.1.2.7 name

```
bannerflow_app.models.ABTest.name = models.CharField(max_length=255, verbose_name="Nazvanie  
testa") [static]
```

7.1.2.8 start_date

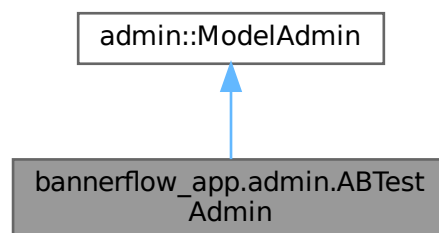
```
bannerflow_app.models.ABTest.start_date = models.DateTimeField(default=timezone.now, verbose←  
_name="Data nachala") [static]
```

The documentation for this class was generated from the following file:

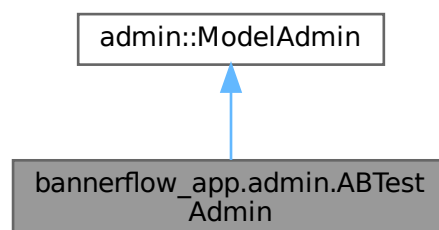
- [bannerflow_app/models.py](#)

7.2 bannerflow_app.admin.ABTestAdmin Class Reference

Inheritance diagram for bannerflow_app.admin.ABTestAdmin:



Collaboration diagram for bannerflow_app.admin.ABTestAdmin:



Static Public Attributes

- tuple `list_display` = ('name', 'campaign', 'audience_segment', 'is_active', 'start_date', 'end_date')
- tuple `list_filter` = ('is_active', 'start_date')
- tuple `search_fields` = ('name', 'campaign__name', 'audience_segment')
- tuple `filter_horizontal` = ('banners',)

7.2.1 Member Data Documentation

7.2.1.1 `filter_horizontal`

```
tuple bannerflow_app.admin.ABTestAdmin.filter_horizontal = ('banners',) [static]
```

7.2.1.2 `list_display`

```
tuple bannerflow_app.admin.ABTestAdmin.list_display = ('name', 'campaign', 'audience_segment', 'is_active', 'start_date', 'end_date') [static]
```

7.2.1.3 `list_filter`

```
tuple bannerflow_app.admin.ABTestAdmin.list_filter = ('is_active', 'start_date') [static]
```

7.2.1.4 `search_fields`

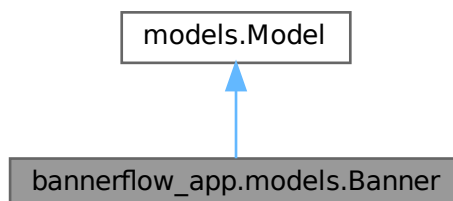
```
tuple bannerflow_app.admin.ABTestAdmin.search_fields = ('name', 'campaign__name', 'audience_↵segment') [static]
```

The documentation for this class was generated from the following file:

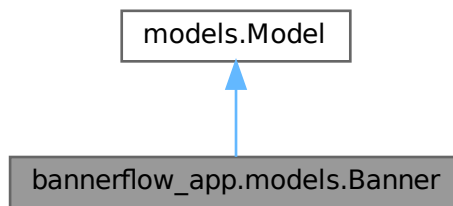
- [bannerflow_app/admin.py](#)

7.3 `bannerflow_app.models.Banner` Class Reference

Inheritance diagram for `bannerflow_app.models.Banner`:



Collaboration diagram for bannerflow_app.models.Banner:



Public Member Functions

- [save](#) (self, *args, **kwargs)
- [clean](#) (self)
- [__str__](#) (self)

Static Public Attributes

- list [MEDIA_TYPE_CHOICES](#)
- [campaign](#) = models.ForeignKey([Campaign](#), on_delete=models.CASCADE, verbose_name="Kampaniya")
- [title](#) = models.CharField(max_length=255, verbose_name="Nazvanie bannera")
- [media_file](#)
- [media_type](#) = models.CharField(max_length=10, choices=[MEDIA_TYPE_CHOICES](#), verbose_name="Tip media")
- [created_at](#) = models.DateTimeField(auto_now_add=True)
- [is_active](#) = models.BooleanField(default=True, verbose_name="Aktiven")

7.3.1 Member Function Documentation

7.3.1.1 `__str__()`

```
bannerflow_app.models.Banner.__str__ (
    self )
```

7.3.1.2 `clean()`

```
bannerflow_app.models.Banner.clean (
    self )
```

Dopolnitelnaya validatsiya pri sohranении

7.3.1.3 save()

```
bannerflow_app.models.Banner.save (
    self,
    * args,
    ** kwargs )
```

7.3.2 Member Data Documentation

7.3.2.1 campaign

```
bannerflow_app.models.Banner.campaign = models.ForeignKey(Campaign, on_delete=models.CASCADE,
verbose_name="Kampaniya") [static]
```

7.3.2.2 created_at

```
bannerflow_app.models.Banner.created_at = models.DateTimeField(auto_now_add=True) [static]
```

7.3.2.3 is_active

```
bannerflow_app.models.Banner.is_active = models.BooleanField(default=True, verbose_name="Aktiven")
[static]
```

7.3.2.4 media_file

```
bannerflow_app.models.Banner.media_file [static]
```

Initial value:

```
= models.FileField(
    upload_to='banners/%Y/%m/%d/',
    verbose_name="Mediafayl",
    help_text="Podderzhivaemye formaty: PNG, JPG, MP4",
    validators=[validate_media_file]
)
```

7.3.2.5 media_type

```
bannerflow_app.models.Banner.media_type = models.CharField(max_length=10, choices=MEDIA_TYPE_CHOICES,
verbose_name="Tip media") [static]
```

7.3.2.6 MEDIA_TYPE_CHOICES

```
list bannerflow_app.models.Banner.MEDIA_TYPE_CHOICES [static]
```

Initial value:

```
= [
    ('image', 'Izobrazhenie'),
    ('video', 'Video'),
]
```

7.3.2.7 title

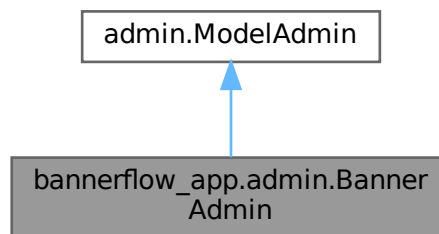
```
bannerflow_app.models.Banner.title = models.CharField(max_length=255, verbose_name="Nazvanie  
bannera") [static]
```

The documentation for this class was generated from the following file:

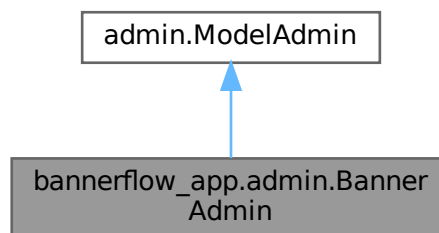
- [bannerflow_app/models.py](#)

7.4 bannerflow_app.admin.BannerAdmin Class Reference

Inheritance diagram for bannerflow_app.admin.BannerAdmin:



Collaboration diagram for bannerflow_app.admin.BannerAdmin:



Static Public Attributes

- tuple `list_display` = ('title', 'campaign', 'media_type', 'is_active', 'created_at')
- tuple `list_filter` = ('media_type', 'is_active', 'created_at')
- tuple `search_fields` = ('title', 'campaign__name')
- tuple `readonly_fields` = ('created_at', 'media_type')

7.4.1 Member Data Documentation

7.4.1.1 list_display

```
tuple bannerflow_app.admin.BannerAdmin.list_display = ('title', 'campaign', 'media_type',  
'is_active', 'created_at') [static]
```

7.4.1.2 list_filter

```
tuple bannerflow_app.admin.BannerAdmin.list_filter = ('media_type', 'is_active', 'created_at')  
[static]
```

7.4.1.3 readonly_fields

```
tuple bannerflow_app.admin.BannerAdmin.readonly_fields = ('created_at', 'media_type') [static]
```

7.4.1.4 search_fields

```
tuple bannerflow_app.admin.BannerAdmin.search_fields = ('title', 'campaign__name') [static]
```

The documentation for this class was generated from the following file:

- bannerflow_app/[admin.py](#)

7.5 bannerflow_app.banner_generator.BannerGenerator Class Reference

Static Public Member Functions

- [generate_html](#) (template, game_data)
- [export_for_godot](#) (html_content, banner_name, template)
- [parse_game_file](#) (file_path)

7.5.1 Detailed Description

Generator interaktivnyh HTML5 bannerov dlya igr

7.5.2 Member Function Documentation

7.5.2.1 export_for_godot()

```
bannerflow_app.banner_generator.BannerGenerator.export_for_godot (  
    html_content,  
    banner_name,  
    template ) [static]
```

Eksport bannera dlya Godot Engine

7.5.2.2 generate_html()

```
bannerflow_app.banner_generator.BannerGenerator.generate_html (
    template,
    game_data ) [static]
```

Generatsiya HTML bannera iz shablona i dannyh igry

7.5.2.3 parse_game_file()

```
bannerflow_app.banner_generator.BannerGenerator.parse_game_file (
    file_path ) [static]
```

Parsing fayla dannyh igry (JSON ili CSV)

The documentation for this class was generated from the following file:

- [bannerflow_app/banner_generator.py](#)

7.6 bannerflow_app.services.BannerService Class Reference

Static Public Member Functions

- [get_active_banners](#) (platform=None)
- [check_budget_and_notify](#) (campaign)

7.6.1 Member Function Documentation

7.6.1.1 check_budget_and_notify()

```
bannerflow_app.services.BannerService.check_budget_and_notify (
    campaign ) [static]
```

Proverit byudzheth kampanii i otpravit uvedomlenie

7.6.1.2 get_active_banners()

```
bannerflow_app.services.BannerService.get_active_banners (
    platform = None ) [static]
```

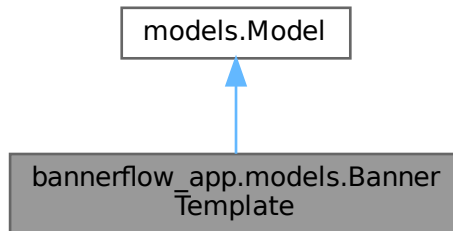
Poluchit aktivnye bannery s filtrom po platforme

The documentation for this class was generated from the following file:

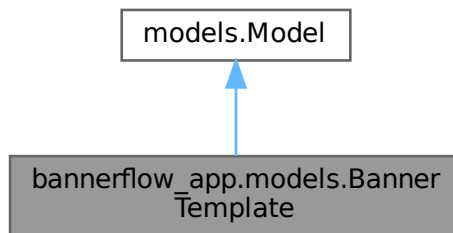
- [bannerflow_app/services.py](#)

7.7 bannerflow_app.models.BannerTemplate Class Reference

Inheritance diagram for bannerflow_app.models.BannerTemplate:



Collaboration diagram for bannerflow_app.models.BannerTemplate:



Classes

- class [Meta](#)

Public Member Functions

- [__str__](#) (self)

Static Public Attributes

- list [TEMPLATE_TYPES](#)
- [name](#) = `models.CharField(max_length=100, verbose_name="Nazvanie shablona")`
- [template_type](#) = `models.CharField(max_length=20, choices=TEMPLATE_TYPES, default='custom', verbose_name="Tip shablona")`

- `html_template` = `models.TextField(verbose_name="HTML шаблон", help_text="Используйте {{ переменные }} для подстановки")`
- `css_styles` = `models.TextField(verbose_name="CSS стили")`
- `javascript` = `models.TextField(verbose_name="JavaScript код", blank=True, help_text="Для интерактивности")`
- `width` = `models.IntegerField(default=300, verbose_name="Ширина (px)")`
- `height` = `models.IntegerField(default=250, verbose_name="Высота (px)")`
- `preview_image` = `models.ImageField(upload_to='templates/previews/', blank=True, verbose_name="Превью")`
- `created_at` = `models.DateTimeField(auto_now_add=True)`

7.7.1 Detailed Description

Шаблон для генерации интерактивных HTML5 баннеров

7.7.2 Member Function Documentation

7.7.2.1 `__str__()`

```
bannerflow_app.models.BannerTemplate.__str__ (
    self )
```

7.7.3 Member Data Documentation

7.7.3.1 `created_at`

```
bannerflow_app.models.BannerTemplate.created_at = models.DateTimeField(auto_now_add=True)
[static]
```

7.7.3.2 `css_styles`

```
bannerflow_app.models.BannerTemplate.css_styles = models.TextField(verbose_name="CSS стили")
[static]
```

7.7.3.3 `height`

```
bannerflow_app.models.BannerTemplate.height = models.IntegerField(default=250, verbose_name="Высота (px)")
[static]
```

7.7.3.4 `html_template`

```
bannerflow_app.models.BannerTemplate.html_template = models.TextField(verbose_name="HTML шаблон",
help_text="Используйте {{ переменные }} для подстановки")
[static]
```

7.7.3.5 `javascript`

```
bannerflow_app.models.BannerTemplate.javascript = models.TextField(verbose_name="JavaScript
код", blank=True, help_text="Для интерактивности")
[static]
```

7.7.3.6 name

```
bannerflow_app.models.BannerTemplate.name = models.CharField(max_length=100, verbose_name="Nazvanie shablona") [static]
```

7.7.3.7 preview_image

```
bannerflow_app.models.BannerTemplate.preview_image = models.ImageField(upload_to='templates/previews/', blank=True, verbose_name="Prevyu") [static]
```

7.7.3.8 template_type

```
bannerflow_app.models.BannerTemplate.template_type = models.CharField(max_length=20, choices=TEMPLATE_TYPES, default='custom', verbose_name="Tip shablona") [static]
```

7.7.3.9 TEMPLATE_TYPES

```
list bannerflow_app.models.BannerTemplate.TEMPLATE_TYPES [static]
```

Initial value:

```
= [
    ('score', 'Banner so schetom'),
    ('achievement', 'Banner s dostizheniem'),
    ('level', 'Banner s urovnem'),
    ('custom', 'Polzovatel'skiy shablon'),
]
```

7.7.3.10 width

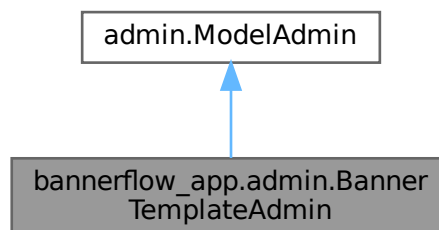
```
bannerflow_app.models.BannerTemplate.width = models.IntegerField(default=300, verbose_name="Shirina (px)") [static]
```

The documentation for this class was generated from the following file:

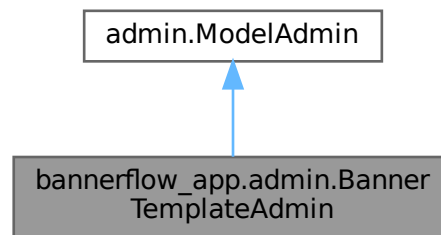
- bannerflow_app/[models.py](#)

7.8 bannerflow_app.admin.BannerTemplateAdmin Class Reference

Inheritance diagram for bannerflow_app.admin.BannerTemplateAdmin:



Collaboration diagram for bannerflow_app.admin.BannerTemplateAdmin:



Static Public Attributes

- tuple `list_display` = ('name', 'template_type', 'width', 'height', 'created_at')
- tuple `list_filter` = ('template_type', 'created_at')
- tuple `search_fields` = ('name', 'html_template')

7.8.1 Member Data Documentation

7.8.1.1 list_display

```
tuple bannerflow_app.admin.BannerTemplateAdmin.list_display = ('name', 'template_type', 'width', 'height', 'created_at') [static]
```

7.8.1.2 list_filter

```
tuple bannerflow_app.admin.BannerTemplateAdmin.list_filter = ('template_type', 'created_at') [static]
```

7.8.1.3 search_fields

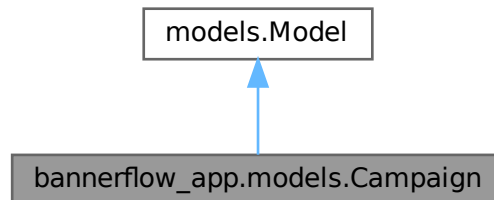
```
tuple bannerflow_app.admin.BannerTemplateAdmin.search_fields = ('name', 'html_template') [static]
```

The documentation for this class was generated from the following file:

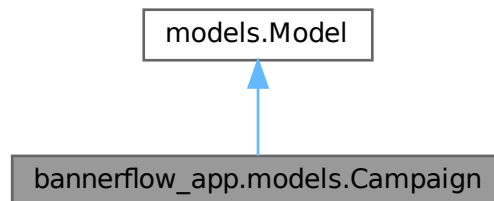
- bannerflow_app/[admin.py](#)

7.9 bannerflow_app.models.Campaign Class Reference

Inheritance diagram for bannerflow_app.models.Campaign:



Collaboration diagram for bannerflow_app.models.Campaign:



Public Member Functions

- [activate](#) (self)
- [pause](#) (self)
- [stop](#) (self)
- [status_color](#) (self)
- [__str__](#) (self)

Static Public Attributes

- list [STATUS_CHOICES](#)
- list [PLATFORM_CHOICES](#)
- [name](#) = models.CharField(max_length=255, verbose_name="Nazvanie kampanii")
- [budget](#)
- [start_date](#) = models.DateTimeField(default=timezone.now, verbose_name="Data nachala")
- [end_date](#) = models.DateTimeField(default=timezone.now, verbose_name="Data okonchaniya")
- [status](#) = models.CharField(max_length=20, choices=[STATUS_CHOICES](#), default='draft', verbose_name="Status")

- [target_geolocation](#)
- [target_platform](#)
- [target_game_version](#)
- [is_ab_test](#) = models.BooleanField(default=False, verbose_name="A/B test")
- [created_by](#) = models.ForeignKey(User, on_delete=models.CASCADE, verbose_name="Sozdate")
- [created_at](#) = models.DateTimeField(auto_now_add=True)

7.9.1 Member Function Documentation

7.9.1.1 `__str__()`

```
bannerflow_app.models.Campaign.__str__ (
    self )
```

7.9.1.2 `activate()`

```
bannerflow_app.models.Campaign.activate (
    self )
```

Aktivirovat kompaniyu

7.9.1.3 `pause()`

```
bannerflow_app.models.Campaign.pause (
    self )
```

Priostanovit kompaniyu

7.9.1.4 `status_color()`

```
bannerflow_app.models.Campaign.status_color (
    self )
```

7.9.1.5 `stop()`

```
bannerflow_app.models.Campaign.stop (
    self )
```

Ostanovit kompaniyu

7.9.2 Member Data Documentation

7.9.2.1 budget

`bannerflow_app.models.Campaign.budget` [static]

Initial value:

```
= models.DecimalField(  
    max_digits=10,  
    decimal_places=2,  
    validators=[MinValueValidator(0)],  
    verbose_name="Byudzhet"  
)
```

7.9.2.2 created_at

`bannerflow_app.models.Campaign.created_at` = `models.DateTimeField(auto_now_add=True)` [static]

7.9.2.3 created_by

`bannerflow_app.models.Campaign.created_by` = `models.ForeignKey(User, on_delete=models.CASCADE, verbose_name="Sozdatel")` [static]

7.9.2.4 end_date

`bannerflow_app.models.Campaign.end_date` = `models.DateTimeField(default=timezone.now, verbose_name="Data okonchaniya")` [static]

7.9.2.5 is_ab_test

`bannerflow_app.models.Campaign.is_ab_test` = `models.BooleanField(default=False, verbose_name="A/B test")` [static]

7.9.2.6 name

`bannerflow_app.models.Campaign.name` = `models.CharField(max_length=255, verbose_name="Nazvanie kampanii")` [static]

7.9.2.7 PLATFORM_CHOICES

`list bannerflow_app.models.Campaign.PLATFORM_CHOICES` [static]

Initial value:

```
= [  
    ('iOS', 'iOS'),  
    ('Android', 'Android'),  
    ('Both', 'iOS i Android'),  
)
```


7.9.2.8 start_date

```
bannerflow_app.models.Campaign.start_date = models.DateTimeField(default=timezone.now, verbose_name="Data nachala") [static]
```

7.9.2.9 status

```
bannerflow_app.models.Campaign.status = models.CharField(max_length=20, choices=STATUS_CHOICES, default='draft', verbose_name="Status") [static]
```

7.9.2.10 STATUS_CHOICES

```
list bannerflow_app.models.Campaign.STATUS_CHOICES [static]
```

Initial value:

```
= [
    ('draft', 'Chernovik'),
    ('active', 'Aktivna'),
    ('paused', 'Priostanovlena'),
    ('stopped', 'Ostanovlena'),
]
```

7.9.2.11 target_game_version

```
bannerflow_app.models.Campaign.target_game_version [static]
```

Initial value:

```
= models.CharField(
    max_length=50,
    blank=True,
    verbose_name="Versiya igry"
)
```

7.9.2.12 target_geolocation

```
bannerflow_app.models.Campaign.target_geolocation [static]
```

Initial value:

```
= models.JSONField(
    default=list,
    verbose_name="Geolokatsiya",
    help_text="Spisok stran/regionov dlya targetinga"
)
```

7.9.2.13 target_platform

```
bannerflow_app.models.Campaign.target_platform [static]
```

Initial value:

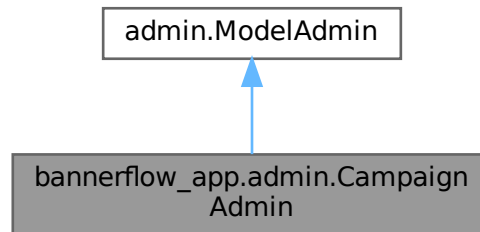
```
= models.CharField(
    max_length=10,
    choices=PLATFORM_CHOICES,
    default='Both',
    verbose_name="Tselevaya platforma"
)
```

The documentation for this class was generated from the following file:

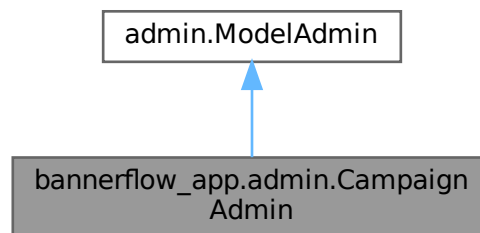
- bannerflow_app/[models.py](#)

7.10 bannerflow_app.admin.CampaignAdmin Class Reference

Inheritance diagram for bannerflow_app.admin.CampaignAdmin:



Collaboration diagram for bannerflow_app.admin.CampaignAdmin:



Public Member Functions

- [activate_campaigns](#) (self, request, queryset)
- [pause_campaigns](#) (self, request, queryset)
- [stop_campaigns](#) (self, request, queryset)

Static Public Attributes

- tuple [list_display](#) = ('name', 'budget', 'start_date', 'end_date', 'status', 'target_platform', 'created_by', 'created_at')
- tuple [list_filter](#) = ('status', 'target_platform', 'is_ab_test', 'created_at')
- tuple [search_fields](#) = ('name', 'target_game_version')
- tuple [readonly_fields](#) = ('created_at',)
- list [actions](#) = [[activate_campaigns](#), [pause_campaigns](#), [stop_campaigns](#)]
- tuple [fieldsets](#)
- [short_description](#)

7.10.1 Member Function Documentation

7.10.1.1 activate_campaigns()

```
bannerflow_app.admin.CampaignAdmin.activate_campaigns (
    self,
    request,
    queryset )
```

7.10.1.2 pause_campaigns()

```
bannerflow_app.admin.CampaignAdmin.pause_campaigns (
    self,
    request,
    queryset )
```

7.10.1.3 stop_campaigns()

```
bannerflow_app.admin.CampaignAdmin.stop_campaigns (
    self,
    request,
    queryset )
```

7.10.2 Member Data Documentation

7.10.2.1 actions

```
list bannerflow_app.admin.CampaignAdmin.actions = ['activate_campaigns', 'pause_campaigns',
'stop_campaigns'] [static]
```

7.10.2.2 fieldsets

```
tuple bannerflow_app.admin.CampaignAdmin.fieldsets [static]
```

Initial value:

```
= (
    ('Osnovnaya informatsiya', {
        'fields': ('name', 'budget', 'created_by')
    }),
    ('Vremennye ramki', {
        'fields': ('start_date', 'end_date')
    }),
    ('Targeting', {
        'fields': ('target_platform', 'target_geolocation', 'target_game_version')
    }),
    ('Status', {
        'fields': ('status', 'is_ab_test')
    }),
)
```

7.10.2.3 list_display

```
tuple bannerflow_app.admin.CampaignAdmin.list_display = ('name', 'budget', 'start_date', 'end←
_date', 'status', 'target_platform', 'created_by', 'created_at') [static]
```

7.10.2.4 list_filter

```
tuple bannerflow_app.admin.CampaignAdmin.list_filter = ('status', 'target_platform', 'is_ab_↵  
test', 'created_at') [static]
```

7.10.2.5 readonly_fields

```
tuple bannerflow_app.admin.CampaignAdmin.readonly_fields = ('created_at',) [static]
```

7.10.2.6 search_fields

```
tuple bannerflow_app.admin.CampaignAdmin.search_fields = ('name', 'target_game_version') [static]
```

7.10.2.7 short_description

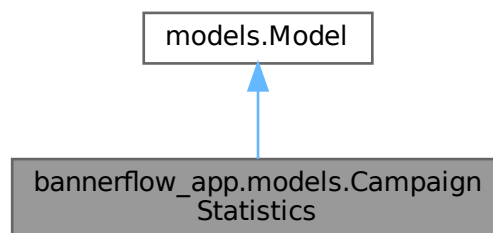
```
bannerflow_app.admin.CampaignAdmin.short_description [static]
```

The documentation for this class was generated from the following file:

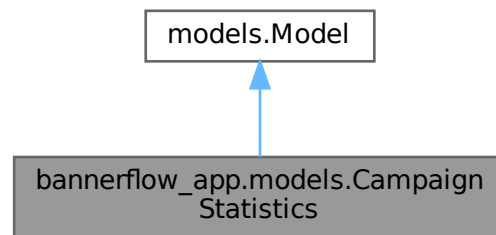
- bannerflow_app/[admin.py](#)

7.11 bannerflow_app.models.CampaignStatistics Class Reference

Inheritance diagram for bannerflow_app.models.CampaignStatistics:



Collaboration diagram for bannerflow_app.models.CampaignStatistics:



Classes

- class [Meta](#)

Public Member Functions

- [ctr](#) (self)
- [__str__](#) (self)

Static Public Attributes

- [campaign](#) = models.ForeignKey([Campaign](#), on_delete=models.CASCADE, verbose_name="Kampaniya")
- [date](#) = models.DateField(default=timezone.now, verbose_name="Data")
- [impressions](#) = models.IntegerField(default=0, verbose_name="Pokazy")
- [clicks](#) = models.IntegerField(default=0, verbose_name="Kliki")
- [conversions](#) = models.IntegerField(default=0, verbose_name="Konversii")

7.11.1 Member Function Documentation

7.11.1.1 [__str__](#)()

```

bannerflow_app.models.CampaignStatistics.__str__ (
    self )

```

7.11.1.2 [ctr](#)()

```

bannerflow_app.models.CampaignStatistics.ctr (
    self )

```

CTR (Click-Through Rate) v protsentah

7.11.2 Member Data Documentation

7.11.2.1 campaign

```
bannerflow_app.models.CampaignStatistics.campaign = models.ForeignKey(Campaign, on_delete=models.CASCADE, verbose_name="Kampaniya") [static]
```

7.11.2.2 clicks

```
bannerflow_app.models.CampaignStatistics.clicks = models.IntegerField(default=0, verbose_name="Kliki") [static]
```

7.11.2.3 conversions

```
bannerflow_app.models.CampaignStatistics.conversions = models.IntegerField(default=0, verbose_name="Konversii") [static]
```

7.11.2.4 date

```
bannerflow_app.models.CampaignStatistics.date = models.DateField(default=timezone.now, verbose_name="Data") [static]
```

7.11.2.5 impressions

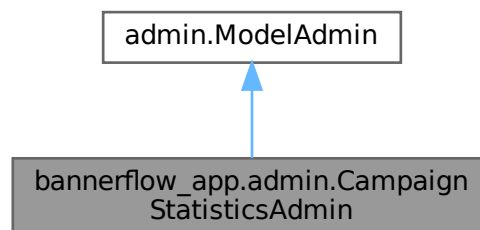
```
bannerflow_app.models.CampaignStatistics.impressions = models.IntegerField(default=0, verbose_name="Pokazy") [static]
```

The documentation for this class was generated from the following file:

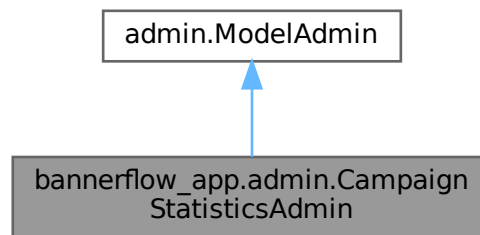
- [bannerflow_app/models.py](#)

7.12 bannerflow_app.admin.CampaignStatisticsAdmin Class Reference

Inheritance diagram for `bannerflow_app.admin.CampaignStatisticsAdmin`:



Collaboration diagram for bannerflow_app.admin.CampaignStatisticsAdmin:



Static Public Attributes

- tuple `list_display` = ('campaign', 'date', 'impressions', 'clicks', 'conversions', 'ctr')
- tuple `list_filter` = ('date', 'campaign')
- tuple `readonly_fields` = ('ctr',)
- tuple `search_fields` = ('campaign__name',)

7.12.1 Member Data Documentation

7.12.1.1 list_display

```
tuple bannerflow_app.admin.CampaignStatisticsAdmin.list_display = ('campaign', 'date', 'impressions', 'clicks', 'conversions', 'ctr') [static]
```

7.12.1.2 list_filter

```
tuple bannerflow_app.admin.CampaignStatisticsAdmin.list_filter = ('date', 'campaign') [static]
```

7.12.1.3 readonly_fields

```
tuple bannerflow_app.admin.CampaignStatisticsAdmin.readonly_fields = ('ctr',) [static]
```

7.12.1.4 search_fields

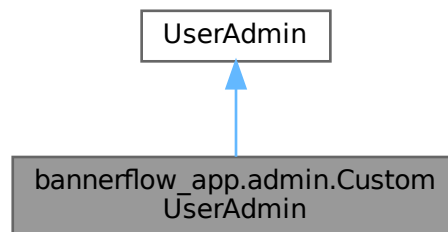
```
tuple bannerflow_app.admin.CampaignStatisticsAdmin.search_fields = ('campaign__name',) [static]
```

The documentation for this class was generated from the following file:

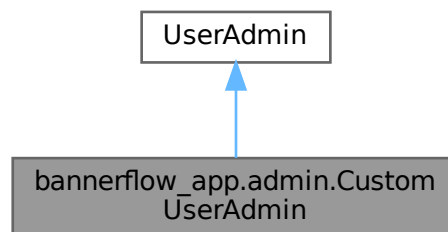
- bannerflow_app/[admin.py](#)

7.13 bannerflow_app.admin.CustomUserAdmin Class Reference

Inheritance diagram for bannerflow_app.admin.CustomUserAdmin:



Collaboration diagram for bannerflow_app.admin.CustomUserAdmin:



Static Public Attributes

- list `inlines` = [`UserProfileInline`]

7.13.1 Member Data Documentation

7.13.1.1 inlines

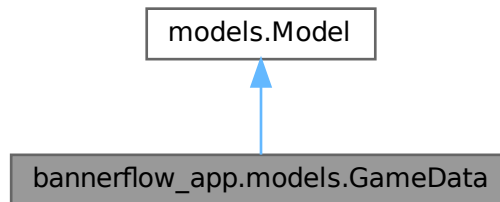
```
list bannerflow_app.admin.CustomUserAdmin.inlines = [UserProfileInline] [static]
```

The documentation for this class was generated from the following file:

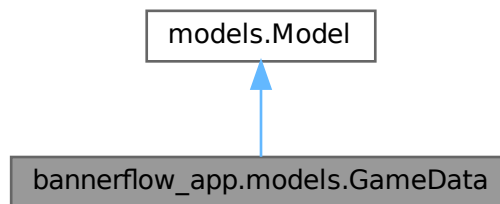
- `bannerflow_app/admin.py`

7.14 bannerflow_app.models.GameData Class Reference

Inheritance diagram for bannerflow_app.models.GameData:



Collaboration diagram for bannerflow_app.models.GameData:



Classes

- class [Meta](#)

Public Member Functions

- [__str__](#) (self)

Static Public Attributes

- list [GAME_TYPES](#)
- [game_name](#) = models.CharField(max_length=100, verbose_name="Nazvanie igry")
- [game_type](#) = models.CharField(max_length=20, choices=[GAME_TYPES](#), default='custom', verbose_name="Tip igry")
- [player_name](#) = models.CharField(max_length=100, verbose_name="Imya igroka")
- [score](#) = models.IntegerField(default=0, verbose_name="Schet")
- [level](#) = models.IntegerField(default=1, verbose_name="Uroven")
- [play_time](#) = models.IntegerField(default=0, verbose_name="Vremya igry (minuty)")
- [achievements](#) = models.JSONField(default=list, verbose_name="Dostizheniya")
- [data_file](#) = models.FileField(upload_to='game_data/', verbose_name="Fayl dannyh igry", help_text="JSON ili CSV fayl")
- [imported_at](#) = models.DateTimeField(auto_now_add=True)

7.14.1 Detailed Description

Dannye iz igry dlya generatsii bannerov

7.14.2 Member Function Documentation

7.14.2.1 `__str__()`

```
bannerflow_app.models.GameData.__str__ (
    self )
```

7.14.3 Member Data Documentation

7.14.3.1 `achievements`

```
bannerflow_app.models.GameData.achievements = models.JSONField(default=list, verbose_name="Dostizheniya")
[static]
```

7.14.3.2 `data_file`

```
bannerflow_app.models.GameData.data_file = models.FileField(upload_to='game_data/', verbose_name="Fayl dannyh igry", help_text="JSON ili CSV fayl") [static]
```

7.14.3.3 `game_name`

```
bannerflow_app.models.GameData.game_name = models.CharField(max_length=100, verbose_name="Nazvanie igry") [static]
```

7.14.3.4 `game_type`

```
bannerflow_app.models.GameData.game_type = models.CharField(max_length=20, choices=GAME_TYPES, default='custom', verbose_name="Tip igry") [static]
```

7.14.3.5 `GAME_TYPES`

```
list bannerflow_app.models.GameData.GAME_TYPES [static]
```

Initial value:

```
= [
    ('platformer', 'Platformer'),
    ('rpg', 'RPG'),
    ('shooter', 'Shuter'),
    ('strategy', 'Strategiya'),
    ('custom', 'Drugaya'),
]
```

7.14.3.6 imported_at

```
bannerflow_app.models.GameData.imported_at = models.DateTimeField(auto_now_add=True) [static]
```

7.14.3.7 level

```
bannerflow_app.models.GameData.level = models.IntegerField(default=1, verbose_name="Uroven")  
[static]
```

7.14.3.8 play_time

```
bannerflow_app.models.GameData.play_time = models.IntegerField(default=0, verbose_name="Vremya  
igry (minuty)") [static]
```

7.14.3.9 player_name

```
bannerflow_app.models.GameData.player_name = models.CharField(max_length=100, verbose_name="Imya  
igroka") [static]
```

7.14.3.10 score

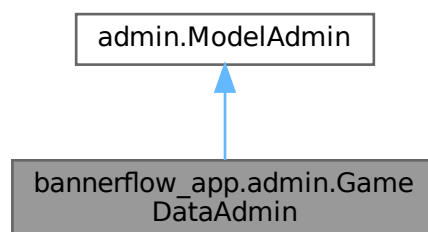
```
bannerflow_app.models.GameData.score = models.IntegerField(default=0, verbose_name="Schet")  
[static]
```

The documentation for this class was generated from the following file:

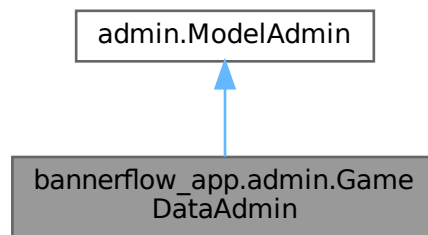
- bannerflow_app/[models.py](#)

7.15 bannerflow_app.admin.GameDataAdmin Class Reference

Inheritance diagram for bannerflow_app.admin.GameDataAdmin:



Collaboration diagram for bannerflow_app.admin.GameDataAdmin:



Static Public Attributes

- tuple `list_display` = ('game_name', 'player_name', 'score', 'level', 'game_type', 'imported_at')
- tuple `list_filter` = ('game_type', 'imported_at')
- tuple `search_fields` = ('game_name', 'player_name')

7.15.1 Member Data Documentation

7.15.1.1 list_display

```
tuple bannerflow_app.admin.GameDataAdmin.list_display = ('game_name', 'player_name', 'score', 'level', 'game_type', 'imported_at') [static]
```

7.15.1.2 list_filter

```
tuple bannerflow_app.admin.GameDataAdmin.list_filter = ('game_type', 'imported_at') [static]
```

7.15.1.3 search_fields

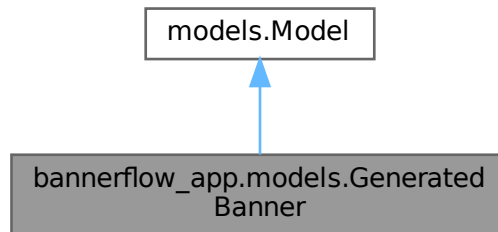
```
tuple bannerflow_app.admin.GameDataAdmin.search_fields = ('game_name', 'player_name') [static]
```

The documentation for this class was generated from the following file:

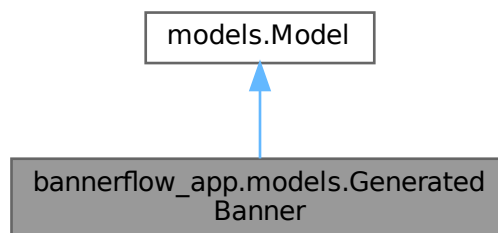
- bannerflow_app/[admin.py](#)

7.16 bannerflow_app.models.GeneratedBanner Class Reference

Inheritance diagram for bannerflow_app.models.GeneratedBanner:



Collaboration diagram for bannerflow_app.models.GeneratedBanner:



Classes

- class [Meta](#)

Public Member Functions

- [__str__](#) (self)

Static Public Attributes

- [name](#) = `models.CharField(max_length=100, verbose_name="Nazvanie bannera")`
- [template](#) = `models.ForeignKey(BannerTemplate, on_delete=models.CASCADE, verbose_name="Shablon")`
- [game_data](#) = `models.ForeignKey(GameData, on_delete=models.CASCADE, verbose_name="Dannye igry")`
- [html_content](#) = `models.TextField(verbose_name="HTML sodержimoe")`
- [godot_export](#) = `models.FileField(upload_to='exports/godot', blank=True, verbose_name="Eksport dlya Godot")`
- [preview_url](#) = `models.URLField(blank=True, verbose_name="URL predprosmotra")`
- [created_at](#) = `models.DateTimeField(auto_now_add=True)`

7.16.1 Detailed Description

Sgenerirovannyi interaktivnyi banner

7.16.2 Member Function Documentation

7.16.2.1 `__str__()`

```
bannerflow_app.models.GeneratedBanner.__str__ (
    self )
```

7.16.3 Member Data Documentation

7.16.3.1 `created_at`

```
bannerflow_app.models.GeneratedBanner.created_at = models.DateTimeField(auto_now_add=True)
[static]
```

7.16.3.2 `game_data`

```
bannerflow_app.models.GeneratedBanner.game_data = models.ForeignKey(GameData, on_delete=models.CASCADE,
verbose_name="Dannye igry") [static]
```

7.16.3.3 `godot_export`

```
bannerflow_app.models.GeneratedBanner.godot_export = models.FileField(upload_to='exports/godot/',
blank=True, verbose_name="Eksport dlya Godot") [static]
```

7.16.3.4 `html_content`

```
bannerflow_app.models.GeneratedBanner.html_content = models.TextField(verbose_name="HTML sodержimoe")
[static]
```

7.16.3.5 `name`

```
bannerflow_app.models.GeneratedBanner.name = models.CharField(max_length=100, verbose_name="Nazvanie
bannera") [static]
```

7.16.3.6 `preview_url`

```
bannerflow_app.models.GeneratedBanner.preview_url = models.URLField(blank=True, verbose_name="URL
predprosmotra") [static]
```

7.16.3.7 template

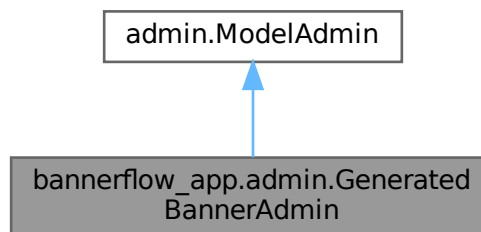
```
bannerflow_app.models.GeneratedBanner.template = models.ForeignKey(BannerTemplate, on_delete=models.CASCADE, verbose_name="Shablon") [static]
```

The documentation for this class was generated from the following file:

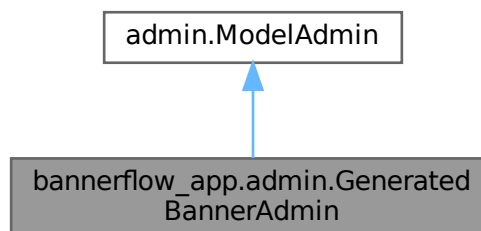
- [bannerflow_app/models.py](#)

7.17 bannerflow_app.admin.GeneratedBannerAdmin Class Reference

Inheritance diagram for bannerflow_app.admin.GeneratedBannerAdmin:



Collaboration diagram for bannerflow_app.admin.GeneratedBannerAdmin:



Static Public Attributes

- tuple `list_display` = ('name', 'template', 'game_data', 'created_at')
- tuple `list_filter` = ('created_at', 'template')
- tuple `search_fields` = ('name', 'html_content')
- tuple `readonly_fields` = ('created_at', 'preview_url')

7.17.1 Member Data Documentation

7.17.1.1 list_display

```
tuple bannerflow_app.admin.GeneratedBannerAdmin.list_display = ('name', 'template', 'game_data', 'created_at') [static]
```

7.17.1.2 list_filter

```
tuple bannerflow_app.admin.GeneratedBannerAdmin.list_filter = ('created_at', 'template') [static]
```

7.17.1.3 readonly_fields

```
tuple bannerflow_app.admin.GeneratedBannerAdmin.readonly_fields = ('created_at', 'preview_url') [static]
```

7.17.1.4 search_fields

```
tuple bannerflow_app.admin.GeneratedBannerAdmin.search_fields = ('name', 'html_content') [static]
```

The documentation for this class was generated from the following file:

- bannerflow_app/[admin.py](#)

7.18 bannerflow_app.models.BannerTemplate.Meta Class Reference

Static Public Attributes

- str [verbose_name](#) = "Shablon bannera"
- str [verbose_name_plural](#) = "Shablony bannerov"

7.18.1 Member Data Documentation

7.18.1.1 verbose_name

```
str bannerflow_app.models.BannerTemplate.Meta.verbose_name = "Shablon bannera" [static]
```

7.18.1.2 verbose_name_plural

```
str bannerflow_app.models.BannerTemplate.Meta.verbose_name_plural = "Shablony bannerov" [static]
```

The documentation for this class was generated from the following file:

- bannerflow_app/[models.py](#)

7.19 bannerflow_app.models.CampaignStatistics.Meta Class Reference

Static Public Attributes

- str `verbose_name` = "Statistika kampanii"
- str `verbose_name_plural` = "Statistika kampaniy"

7.19.1 Member Data Documentation

7.19.1.1 `verbose_name`

```
str bannerflow_app.models.CampaignStatistics.Meta.verbose_name = "Statistika kampanii" [static]
```

7.19.1.2 `verbose_name_plural`

```
str bannerflow_app.models.CampaignStatistics.Meta.verbose_name_plural = "Statistika kampaniy" [static]
```

The documentation for this class was generated from the following file:

- bannerflow_app/[models.py](#)

7.20 bannerflow_app.models.GameData.Meta Class Reference

Static Public Attributes

- str `verbose_name` = "Dannye igry"
- str `verbose_name_plural` = "Dannye igr"

7.20.1 Member Data Documentation

7.20.1.1 `verbose_name`

```
str bannerflow_app.models.GameData.Meta.verbose_name = "Dannye igry" [static]
```

7.20.1.2 `verbose_name_plural`

```
str bannerflow_app.models.GameData.Meta.verbose_name_plural = "Dannye igr" [static]
```

The documentation for this class was generated from the following file:

- bannerflow_app/[models.py](#)

7.21 bannerflow_app.models.GeneratedBanner.Meta Class Reference

Static Public Attributes

- str [verbose_name](#) = "Sgenerirovanny banner"
- str [verbose_name_plural](#) = "Sgenerirovannye bannery"

7.21.1 Member Data Documentation

7.21.1.1 verbose_name

```
str bannerflow_app.models.GeneratedBanner.Meta.verbose_name = "Sgenerirovanny banner" [static]
```

7.21.1.2 verbose_name_plural

```
str bannerflow_app.models.GeneratedBanner.Meta.verbose_name_plural = "Sgenerirovannye bannery" [static]
```

The documentation for this class was generated from the following file:

- bannerflow_app/[models.py](#)

7.22 bannerflow_app.services.NotificationService Class Reference

Static Public Member Functions

- [send_budget_notification](#) (campaign)

7.22.1 Detailed Description

Servis otpravki uvedomleniy

7.22.2 Member Function Documentation

7.22.2.1 send_budget_notification()

```
bannerflow_app.services.NotificationService.send_budget_notification (
    campaign ) [static]
```

Otpravka uvedomleniya ob ischerpanii byudzheta

The documentation for this class was generated from the following file:

- bannerflow_app/[services.py](#)

7.23 bannerflow_app.services.StatisticsService Class Reference

Static Public Member Functions

- [calculate_ctr](#) (clicks, impressions)
- [get_campaign_stats](#) (campaign)

7.23.1 Detailed Description

Servis raboty so statistikoy

7.23.2 Member Function Documentation

7.23.2.1 calculate_ctr()

```
bannerflow_app.services.StatisticsService.calculate_ctr (  
    clicks,  
    impressions ) [static]
```

Raschet CTR (Click-Through Rate)

7.23.2.2 get_campaign_stats()

```
bannerflow_app.services.StatisticsService.get_campaign_stats (  
    campaign ) [static]
```

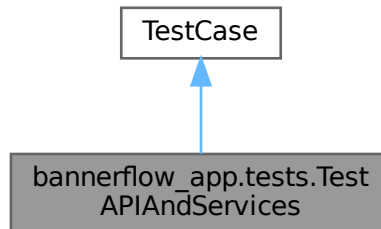
Poluchenie statistiki po kampanii

The documentation for this class was generated from the following file:

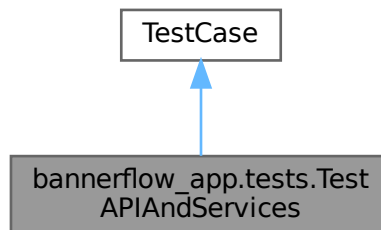
- [bannerflow_app/services.py](#)

7.24 bannerflow_app.tests.TestAPIAndServices Class Reference

Inheritance diagram for bannerflow_app.tests.TestAPIAndServices:



Collaboration diagram for bannerflow_app.tests.TestAPIAndServices:



Public Member Functions

- [setUp](#) (self)
- [test_home_page](#) (self)
- [test_api_endpoint](#) (self)
- [test_services](#) (self)

Public Attributes

- [client](#)
- [user](#)
- [campaign](#)
- [banner](#)

7.24.1 Detailed Description

Testirovanie API i servisov

7.24.2 Member Function Documentation

7.24.2.1 setUp()

```
bannerflow_app.tests.TestAPIAndServices.setUp (  
    self )
```

7.24.2.2 test_api_endpoint()

```
bannerflow_app.tests.TestAPIAndServices.test_api_endpoint (  
    self )
```

Test API endpoints

7.24.2.3 test_home_page()

```
bannerflow_app.tests.TestAPIAndServices.test_home_page (  
    self )
```

Test glavnoy stranitsy

7.24.2.4 test_services()

```
bannerflow_app.tests.TestAPIAndServices.test_services (  
    self )
```

Test servisov

7.24.3 Member Data Documentation

7.24.3.1 banner

```
bannerflow_app.tests.TestAPIAndServices.banner
```

7.24.3.2 campaign

```
bannerflow_app.tests.TestAPIAndServices.campaign
```

7.24.3.3 client

`bannerflow_app.tests.TestAPIAndServices.client`

7.24.3.4 user

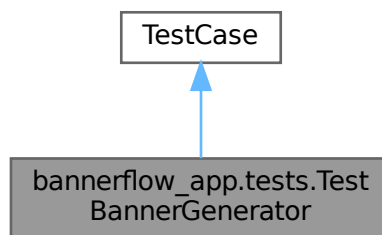
`bannerflow_app.tests.TestAPIAndServices.user`

The documentation for this class was generated from the following file:

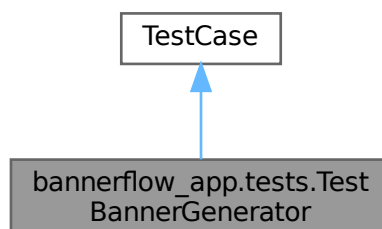
- `bannerflow_app/`[tests.py](#)

7.25 bannerflow_app.tests.TestBannerGenerator Class Reference

Inheritance diagram for `bannerflow_app.tests.TestBannerGenerator`:



Collaboration diagram for `bannerflow_app.tests.TestBannerGenerator`:



Public Member Functions

- [setUp](#) (self)
- [test_banner_generation](#) (self)
- [test_game_file_parsing](#) (self)

Public Attributes

- [template](#)
- [game_data](#)

7.25.1 Detailed Description

Testirovanie generatora bannerov

7.25.2 Member Function Documentation

7.25.2.1 setUp()

```
bannerflow_app.tests.TestBannerGenerator.setUp (  
    self )
```

7.25.2.2 test_banner_generation()

```
bannerflow_app.tests.TestBannerGenerator.test_banner_generation (  
    self )
```

Test generatsii HTML bannera

7.25.2.3 test_game_file_parsing()

```
bannerflow_app.tests.TestBannerGenerator.test_game_file_parsing (  
    self )
```

Test parsinga faylov igry

7.25.3 Member Data Documentation

7.25.3.1 game_data

```
bannerflow_app.tests.TestBannerGenerator.game_data
```

7.25.3.2 `template`

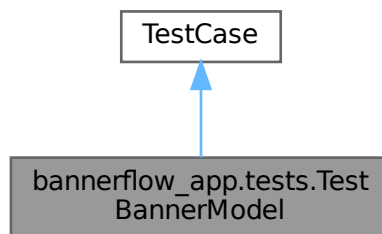
`bannerflow_app.tests.TestBannerGenerator.template`

The documentation for this class was generated from the following file:

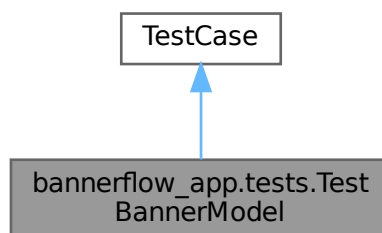
- `bannerflow_app/tests.py`

7.26 `bannerflow_app.tests.TestBannerModel` Class Reference

Inheritance diagram for `bannerflow_app.tests.TestBannerModel`:



Collaboration diagram for `bannerflow_app.tests.TestBannerModel`:



Public Member Functions

- `setUp` (self)
- `test_banner_creation` (self)

Public Attributes

- [user](#)
- [campaign](#)

7.26.1 Detailed Description

Testirovanie modeli Banner

7.26.2 Member Function Documentation

7.26.2.1 setUp()

```
bannerflow_app.tests.TestBannerModel.setUp (
    self )
```

7.26.2.2 test_banner_creation()

```
bannerflow_app.tests.TestBannerModel.test_banner_creation (
    self )
```

Test sozdaniya bannera

7.26.3 Member Data Documentation

7.26.3.1 campaign

```
bannerflow_app.tests.TestBannerModel.campaign
```

7.26.3.2 user

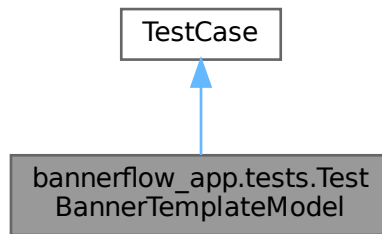
```
bannerflow_app.tests.TestBannerModel.user
```

The documentation for this class was generated from the following file:

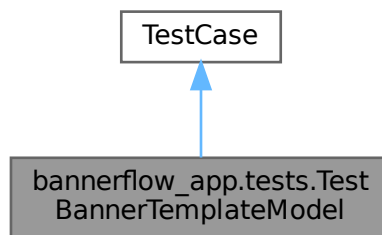
- [bannerflow_app/tests.py](#)

7.27 bannerflow_app.tests.TestBannerTemplateModel Class Reference

Inheritance diagram for bannerflow_app.tests.TestBannerTemplateModel:



Collaboration diagram for bannerflow_app.tests.TestBannerTemplateModel:



Public Member Functions

- [test_template_creation](#) (self)

7.27.1 Detailed Description

Testirovanie modeli BannerTemplate

7.27.2 Member Function Documentation

7.27.2.1 test_template_creation()

```
bannerflow_app.tests.TestBannerTemplateModel.test_template_creation (
    self )
```

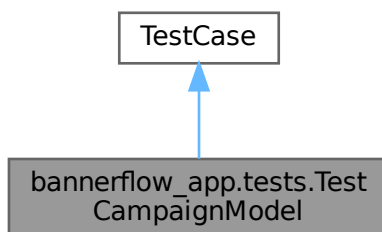
Test sozdaniya shablona bannera

The documentation for this class was generated from the following file:

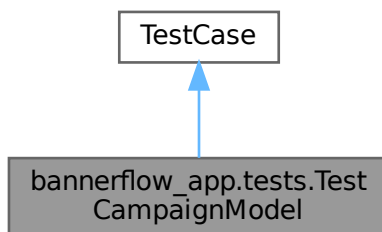
- bannerflow_app/[tests.py](#)

7.28 bannerflow_app.tests.TestCampaignModel Class Reference

Inheritance diagram for bannerflow_app.tests.TestCampaignModel:



Collaboration diagram for bannerflow_app.tests.TestCampaignModel:



Public Member Functions

- [setUp](#) (self)
- [test_campaign_creation_and_methods](#) (self)

Public Attributes

- [user](#)

7.28.1 Detailed Description

Testirovanie modeli Campaign

7.28.2 Member Function Documentation

7.28.2.1 setUp()

```
bannerflow_app.tests.TestCampaignModel.setUp (
    self )
```

7.28.2.2 test_campaign_creation_and_methods()

```
bannerflow_app.tests.TestCampaignModel.test_campaign_creation_and_methods (
    self )
```

Test sozdaniya kampanii i ee metodov

7.28.3 Member Data Documentation

7.28.3.1 user

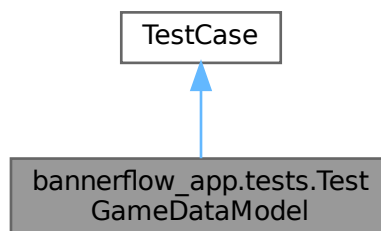
```
bannerflow_app.tests.TestCampaignModel.user
```

The documentation for this class was generated from the following file:

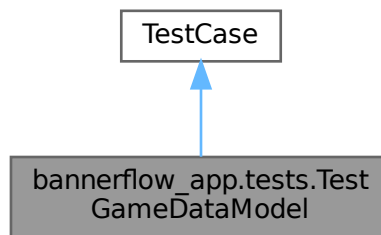
- bannerflow_app/[tests.py](#)

7.29 bannerflow_app.tests.TestGameDataModel Class Reference

Inheritance diagram for bannerflow_app.tests.TestGameDataModel:



Collaboration diagram for bannerflow_app.tests.TestGameDataModel:



Public Member Functions

- [test_game_data_creation](#) (self)

7.29.1 Detailed Description

Testirovanie modeli GameData

7.29.2 Member Function Documentation

7.29.2.1 test_game_data_creation()

```
bannerflow_app.tests.TestGameDataModel.test_game_data_creation (
    self )
```

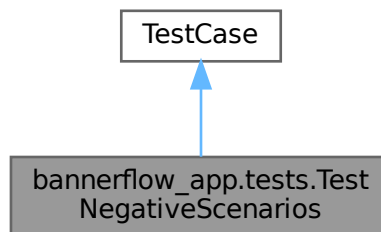
Test sozdaniya dannyh igry

The documentation for this class was generated from the following file:

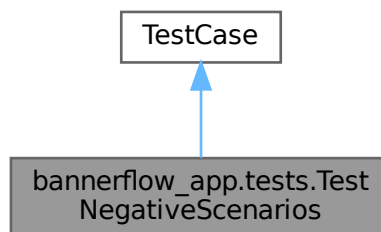
- [bannerflow_app/tests.py](#)

7.30 bannerflow_app.tests.TestNegativeScenarios Class Reference

Inheritance diagram for bannerflow_app.tests.TestNegativeScenarios:



Collaboration diagram for bannerflow_app.tests.TestNegativeScenarios:



Public Member Functions

- [setUp](#) (self)
- [test_negative_cases](#) (self)

Public Attributes

- [user](#)

7.30.1 Detailed Description

Testirovanie oshibochnyh stsensariiev

7.30.2 Member Function Documentation

7.30.2.1 setUp()

```
bannerflow_app.tests.TestNegativeScenarios.setUp (
    self )
```

7.30.2.2 test_negative_cases()

```
bannerflow_app.tests.TestNegativeScenarios.test_negative_cases (
    self )
```

Testirovanie oshibochnyh stsensariev

7.30.3 Member Data Documentation

7.30.3.1 user

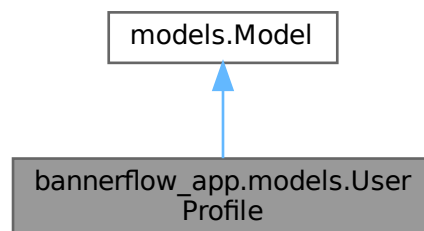
```
bannerflow_app.tests.TestNegativeScenarios.user
```

The documentation for this class was generated from the following file:

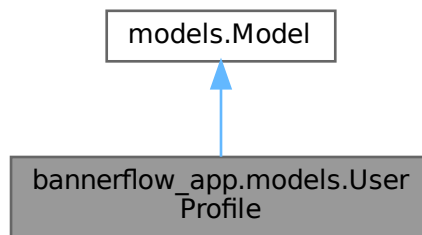
- bannerflow_app/[tests.py](#)

7.31 bannerflow_app.models.UserProfile Class Reference

Inheritance diagram for bannerflow_app.models.UserProfile:



Collaboration diagram for bannerflow_app.models.UserProfile:



Public Member Functions

- `__str__` (self)

Static Public Attributes

- list `ROLE_CHOICES`
- `user` = `models.OneToOneField(User, on_delete=models.CASCADE)`
- `role` = `models.CharField(max_length=20, choices=ROLE_CHOICES, default='marketer')`
- `phone` = `models.CharField(max_length=20, blank=True)`
- `created_at` = `models.DateTimeField(auto_now_add=True)`

7.31.1 Member Function Documentation

7.31.1.1 `__str__()`

```

bannerflow_app.models.UserProfile.__str__ (
    self )

```

7.31.2 Member Data Documentation

7.31.2.1 `created_at`

```

bannerflow_app.models.UserProfile.created_at = models.DateTimeField(auto_now_add=True) [static]

```

7.31.2.2 `phone`

```

bannerflow_app.models.UserProfile.phone = models.CharField(max_length=20, blank=True) [static]

```


7.31.2.3 role

```
bannerflow_app.models.UserProfile.role = models.CharField(max_length=20, choices=ROLE_CHOICES,  
default='marketer') [static]
```

7.31.2.4 ROLE_CHOICES

```
list bannerflow_app.models.UserProfile.ROLE_CHOICES [static]
```

Initial value:

```
= [  
    ('marketer', 'Marketolog'),  
    ('analyst', 'Analitik'),  
    ('admin', 'Administrator'),  
]
```

7.31.2.5 user

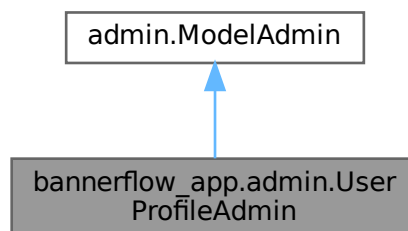
```
bannerflow_app.models.UserProfile.user = models.OneToOneField(User, on_delete=models.CASCADE)  
[static]
```

The documentation for this class was generated from the following file:

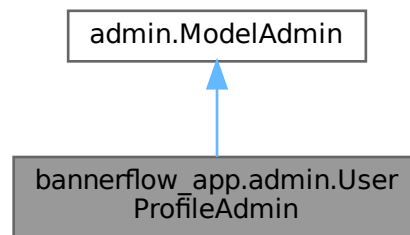
- bannerflow_app/[models.py](#)

7.32 bannerflow_app.admin.UserProfileAdmin Class Reference

Inheritance diagram for bannerflow_app.admin.UserProfileAdmin:



Collaboration diagram for bannerflow_app.admin.UserProfileAdmin:



Static Public Attributes

- tuple `list_display` = ('user', 'role', 'phone', 'created_at')
- tuple `list_filter` = ('role', 'created_at')
- tuple `search_fields` = ('user__username', 'user__email', 'phone')

7.32.1 Member Data Documentation

7.32.1.1 list_display

```
tuple bannerflow_app.admin.UserProfileAdmin.list_display = ('user', 'role', 'phone', 'created_at') [static]
```

7.32.1.2 list_filter

```
tuple bannerflow_app.admin.UserProfileAdmin.list_filter = ('role', 'created_at') [static]
```

7.32.1.3 search_fields

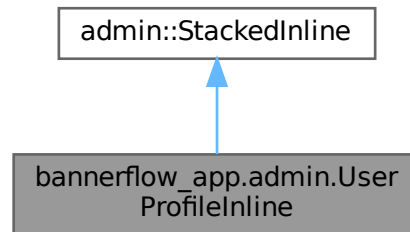
```
tuple bannerflow_app.admin.UserProfileAdmin.search_fields = ('user__username', 'user__email', 'phone') [static]
```

The documentation for this class was generated from the following file:

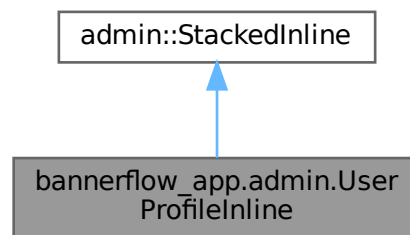
- bannerflow_app/[admin.py](#)

7.33 bannerflow_app.admin.UserProfileInline Class Reference

Inheritance diagram for bannerflow_app.admin.UserProfileInline:



Collaboration diagram for bannerflow_app.admin.UserProfileInline:



Static Public Attributes

- `model` = `UserProfile`
- `bool can_delete` = `False`

7.33.1 Member Data Documentation

7.33.1.1 `can_delete`

```
bool bannerflow_app.admin.UserProfileInline.can_delete = False [static]
```

7.33.1.2 `model`

```
bannerflow_app.admin.UserProfileInline.model = UserProfile [static]
```

The documentation for this class was generated from the following file:

- `bannerflow_app/admin.py`

Chapter 8

File Documentation

8.1 bannerflow_app/admin.py File Reference

Classes

- class [bannerflow_app.admin.UserProfileInline](#)
- class [bannerflow_app.admin.CustomUserAdmin](#)
- class [bannerflow_app.admin.UserProfileAdmin](#)
- class [bannerflow_app.admin.CampaignAdmin](#)
- class [bannerflow_app.admin.BannerAdmin](#)
- class [bannerflow_app.admin.ABTestAdmin](#)
- class [bannerflow_app.admin.CampaignStatisticsAdmin](#)
- class [bannerflow_app.admin.BannerTemplateAdmin](#)
- class [bannerflow_app.admin.GameDataAdmin](#)
- class [bannerflow_app.admin.GeneratedBannerAdmin](#)

Namespaces

- namespace [bannerflow_app](#)
- namespace [bannerflow_app.admin](#)

8.2 bannerflow_app/banner_generator.py File Reference

Classes

- class [bannerflow_app.banner_generator.BannerGenerator](#)

Namespaces

- namespace [bannerflow_app](#)
- namespace [bannerflow_app.banner_generator](#)

8.3 bannerflow_app/models.py File Reference

Classes

- class [bannerflow_app.models.UserProfile](#)
- class [bannerflow_app.models.Campaign](#)
- class [bannerflow_app.models.Banner](#)
- class [bannerflow_app.models.ABTest](#)
- class [bannerflow_app.models.CampaignStatistics](#)
- class [bannerflow_app.models.CampaignStatistics.Meta](#)
- class [bannerflow_app.models.BannerTemplate](#)
- class [bannerflow_app.models.BannerTemplate.Meta](#)
- class [bannerflow_app.models.GameData](#)
- class [bannerflow_app.models.GameData.Meta](#)
- class [bannerflow_app.models.GeneratedBanner](#)
- class [bannerflow_app.models.GeneratedBanner.Meta](#)

Namespaces

- namespace [bannerflow_app](#)
- namespace [bannerflow_app.models](#)

Functions

- [bannerflow_app.models.validate_media_file](#) (value)

8.4 bannerflow_app/services.py File Reference

Classes

- class [bannerflow_app.services.BannerService](#)
- class [bannerflow_app.services.NotificationService](#)
- class [bannerflow_app.services.StatisticsService](#)

Namespaces

- namespace [bannerflow_app](#)
- namespace [bannerflow_app.services](#)

8.5 bannerflow_app/tests.py File Reference

Classes

- class [bannerflow_app.tests.TestCampaignModel](#)
- class [bannerflow_app.tests.TestBannerModel](#)
- class [bannerflow_app.tests.TestBannerTemplateModel](#)
- class [bannerflow_app.tests.TestGameDataModel](#)
- class [bannerflow_app.tests.TestBannerGenerator](#)
- class [bannerflow_app.tests.TestNegativeScenarios](#)
- class [bannerflow_app.tests.TestAPIAndServices](#)

Namespaces

- namespace [bannerflow_app](#)
- namespace [bannerflow_app.tests](#)

8.6 bannerflow_app/views.py File Reference

Namespaces

- namespace [bannerflow_app](#)
- namespace [bannerflow_app.views](#)

Functions

- [bannerflow_app.views.home](#) (request)
- [bannerflow_app.views.dashboard](#) (request)
- [bannerflow_app.views.banner_generator](#) (request)
- [bannerflow_app.views.generate_banner](#) (request)
- [bannerflow_app.views.banner_preview](#) (request, banner_id)
- [bannerflow_app.views.api_banners](#) (request)
- [bannerflow_app.views.upload_game_data](#) (request)
- [bannerflow_app.views.export_banner](#) (request, banner_id)
- [bannerflow_app.views.banner_list](#) (request)

8.7 bannerflow_app/__init__.py File Reference

Namespaces

- namespace [bannerflow_app](#)

8.8 bannerflow_project/__init__.py File Reference

Namespaces

- namespace [bannerflow_project](#)

8.9 bannerflow_project/settings.py File Reference

Namespaces

- namespace [bannerflow_project](#)
- namespace [bannerflow_project.settings](#)

Variables

- `bannerflow_project.settings.BASE_DIR` = `Path(__file__).resolve().parent.parent`
- `str bannerflow_project.settings.SECRET_KEY` = `'django-insecure-your-secret-key-here'`
- `bool bannerflow_project.settings.DEBUG` = `True`
- `list bannerflow_project.settings.ALLOWED_HOSTS` = `[]`
- `list bannerflow_project.settings.INSTALLED_APPS`
- `list bannerflow_project.settings.MIDDLEWARE`
- `str bannerflow_project.settings.ROOT_URLCONF` = `'bannerflow_project.urls'`
- `list bannerflow_project.settings.TEMPLATES`
- `str bannerflow_project.settings.WSGI_APPLICATION` = `'bannerflow_project.wsgi.application'`
- `dict bannerflow_project.settings.DATABASES`
- `str bannerflow_project.settings.LANGUAGE_CODE` = `'en-us'`
- `str bannerflow_project.settings.TIME_ZONE` = `'UTC'`
- `bool bannerflow_project.settings.USE_I18N` = `True`
- `bool bannerflow_project.settings.USE_TZ` = `True`
- `str bannerflow_project.settings.STATIC_URL` = `'static/'`
- `str bannerflow_project.settings.DEFAULT_AUTO_FIELD` = `'django.db.models.BigAutoField'`
- `dict bannerflow_project.settings.REST_FRAMEWORK`
- `str bannerflow_project.settings.MEDIA_URL` = `'/media/'`
- `bannerflow_project.settings.MEDIA_ROOT` = `os.path.join(BASE_DIR, 'media')`
- `str bannerflow_project.settings.EMAIL_BACKEND` = `'django.core.mail.backends.console.EmailBackend'`
- `str bannerflow_project.settings.DEFAULT_FROM_EMAIL` = `'noreply@bannerflow.com'`
- `str bannerflow_project.settings.EMAIL_HOST` = `'smtp.mail.ru'`
- `int bannerflow_project.settings.EMAIL_PORT` = `465`
- `bool bannerflow_project.settings.EMAIL_USE_SSL` = `True`
- `str bannerflow_project.settings.EMAIL_HOST_USER` = `'bannerflow_manager@mail.ru'`
- `str bannerflow_project.settings.EMAIL_HOST_PASSWORD` = `'DyAQzqEBGdfyyBBI3iym'`
- `str bannerflow_project.settings.SERVER_EMAIL` = `'bannerflow_manager@mail.ru'`

8.10 bannerflow_app/urls.py File Reference

Namespaces

- namespace `bannerflow_app`
- namespace `bannerflow_app.urls`

Variables

- `list bannerflow_app.urls.urlpatterns`

8.11 bannerflow_project/urls.py File Reference

Namespaces

- namespace `bannerflow_project`
- namespace `bannerflow_project.urls`

Variables

- list [bannerflow_project.urls.urlpatterns](#)
- [bannerflow_project.urls.MEDIA_URL](#)
- [bannerflow_project.urls.document_root](#)

8.12 bannerflow_project/wsgi.py File Reference

Namespaces

- namespace [bannerflow_project](#)
- namespace [bannerflow_project.wsgi](#)

Variables

- [bannerflow_project.wsgi.application](#) = `get_wsgi_application()`

8.13 manage.py File Reference

Namespaces

- namespace [manage](#)

Functions

- [manage.main](#) ()

8.14 README.md File Reference

